EXPVID: A BENCHMARK FOR EXPERIMENT VIDEO UNDERSTANDING & REASONING

Taro Yamada

ABSTRACT

We introduce ExpVid, a benchmark for evaluating multimodal large language models (MLLMs) on authentic wet-lab experiment videos. Existing video and multimodal evaluations emphasize generic actions, short horizons, or knowledge questions that can often be answered from text alone, failing to test whether systems can perceive fine-grained visual details, track long procedures, and reach evidence-based scientific conclusions. ExpVid addresses this gap with a three-level hierarchy that mirrors the scientific workflow: Level-1 assesses vision-centric recognition of tools, materials, quantities, and operations on 8 s clips; Level-2 probes procedural fidelity via ordering, sequence generation, completeness verification, and next-step prediction on 48 s stage segments; Level-3 evaluates full-experiment scientific analysis via cloze questions requiring integration of evidence across the entire video. The benchmark comprises 7,800 QA pairs from 390 peer-reviewed JoVE experiments across 13 disciplines (mean 489 s per video), constructed by a semi-automatic, vision-centric pipeline with expert validation. We evaluate 19 open- and closed-source MLLMs under both "think" and non-think modes using task-appropriate metrics (Top-1 accuracy, Jaccard, and per-blank accuracy with an automatic judge). Results show substantial headroom: closed-source models lead on average, with GPT-5 strongest at long-horizon procedure and reasoning, while the best open-source model lags notably; models are asymmetric across procedural skills and "thinking" does not uniformly help, suggesting reasoning drift from visuals. Ablations indicate strong dependence on visual evidence, especially for long-horizon reasoning. ExpVid enables more reliable and diagnostic evaluation of vision-grounded experimental competence and highlights key gaps in procedural prediction and long-context scientific reasoning.

1 Introduction

Reliable evaluation of multimodal systems requires tests that reflect the structure and demands of real tasks. Current video and multimodal benchmarks largely emphasize short actions, narrated tasks, or knowledge questions that can be answered without deep visual grounding, limiting their diagnostic power on long-horizon scientific activities. In wet-lab experiments, meaningful progress depends on fine-grained perception (e.g., correctly identifying instruments and quantities), procedural fidelity (e.g., ordering and completeness of steps), and the ability to synthesize evidence into sound conclusions. Existing resources provide valuable coverage of general video understanding or multi-discipline knowledge [Li et al., 2024, Fu et al., 2025, Zhou et al., 2024, Yu et al., 2025, Yue et al., 2024, Chen et al., 2024], but do not simultaneously require vision-centric grounding, long-horizon procedure tracking, and end-to-end scientific reasoning on authentic laboratory workflows.

We present ExpVid, a benchmark designed to evaluate MLLMs in authentic wet-lab settings using peer-reviewed experiment videos from JoVE [JoVE]. ExpVidmirrors the scientific workflow via a three-level hierarchy: Level-1 focuses on perception in short clips; Level-2 probes procedural competence in stage segments; Level-3 tests full-experiment reasoning. The design emphasizes vision-centric annotations to limit shortcutting via text priors and to better isolate the visual and procedural capabilities of models.

Our main contributions are: - A benchmark targeting authentic experiment videos with a three-level hierarchy from fine-grained perception to scientific reasoning, spanning 7,800 QA pairs over 390 JoVE experiments in 13 disciplines. - A semi-automatic, vision-centric construction pipeline that combines LLM assistance with expert validation to curate diverse tasks and control for text-only shortcuts. - A rigorous evaluation protocol with task-appropriate metrics and a comprehensive study of 19 MLLMs under "think" and non-think modes, revealing asymmetric procedural skills, long-horizon challenges, and mixed effects of thinking.

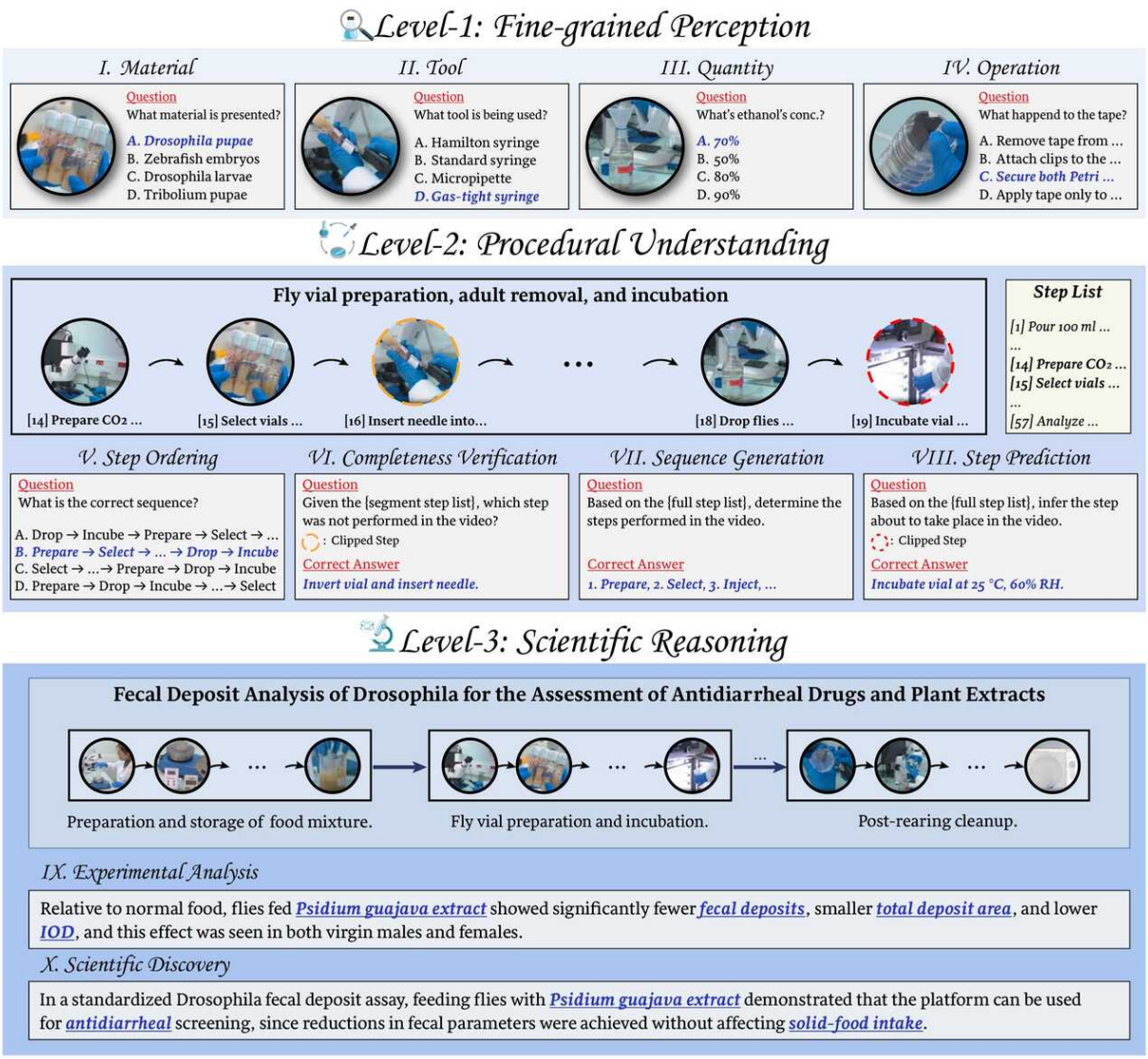

Figure 1: Three-level task hierarchy in ExpVid. Level-1: vision-centric recognition on short clips; Level-2: procedural evaluation on stage segments; Level-3: full-experiment scientific reasoning.

Figure 1 overviews the task hierarchy and Figure 2 summarizes the construction pipeline. We compare ExpVidwith prior benchmarks in Table 1 and present model results in Table 2.

2 Related Work

General video understanding. Benchmarks such as MVBench [Li et al., 2024], Video-MME [Fu et al., 2025], MLVU [Zhou et al., 2024], and VRBench [Yu et al., 2025] have advanced evaluation of temporal comprehension, narrative understanding, and multi-modal inputs. They cover diverse durations and domains, yet typically do not target the vision-grounded, stepwise demands of laboratory procedures.

Knowledge-driven multimodal evaluation. Multi-discipline and science-focused benchmarks (e.g., MMMU [Yue et al., 2024], MMVU [Zhao et al., 2025], Video-MMMU [Hu et al., 2025], ScienceQA [Saikh et al., 2022], SciBench [Wang et al., 2023], MathVista [Lu et al., 2023], MathVision [Wang et al., 2024], MMStar [Chen et al., 2024], SFE [Zhou et al., 2025]) surface reasoning with domain knowledge and visual content. However, many rely on static images, synthetic questions, or short clips, and rarely require tracking experimental procedures end-to-end.

Table 1: Comparison between ExpVid and some MLLM benchmarks. A and M indicate automatic and manual annotation, respectively.

Benchmark	#QA Pairs	#Videos	Avg. Sec.	#Task Types	Annotation	Domain
General Video Benchmarks						
MVBench [Li et al., 2024]	4,000	3,641	16.0	20	A+M	General
Video-MME [Fu et al., 2025]	2,700	900	1,017.9	1	M	General
MLVU [Zhou et al., 2024]	3,102	1,730	930.0	9	M	Narrative
VRBench [Yu et al., 2025]	9,468	960	5,796.0	1	M	Narrative
Knowledge-driven Benchmarks						
MMVU [Zhao et al., 2025]	3,000	1,529	51.4	2	M	Multi-disc.
Video-MMMU [Hu et al., 2025]	900	300	506.2	3	M	Multi-disc.
MathVision [Wang et al., 2024]	3,040	–	–	1	M	Math
MathVista [Lu et al., 2023]	6,141	–	–	31	M	Math
MMMU [Yue et al., 2024]	11,500	–	–	2	M	Multi-disc.
ScienceQA [Saikh et al., 2022]	21,208	–	–	1	M	Science
SciBench [Wang et al., 2023]	789	–	–	1	M	Science
MMStar [Chen et al., 2024]	1,500	–	–	6	M	Multi-disc.
SFE [Zhou et al., 2025]	830	–	–	66	M	Science
ExpVid	7,800	390	489.0	10	A+M	Science

Positioning of ExpVid. As summarized in Table 1, ExpVidcomplements prior work by focusing specifically on authentic laboratory videos from JoVE [JoVE] with vision-centric annotations designed to avoid answerability from text alone, and by structuring tasks to probe perception, procedure, and reasoning over increasing temporal scales.

3 Benchmark Design

Target capability. ExpVidevaluates whether MLLMs can perceive visually grounded experimental details, follow procedures, and draw valid scientific conclusions from authentic wet-lab videos.

Task hierarchy and formats. ExpVidcomprises 10 tasks organized into three levels (Figure 1): - Level-1 (clip-level, 8 s): Multiple-choice recognition of Tools, Materials, Quantities, and Operations, emphasizing visual grounding with plausible distractors. - Level-2 (stage-level, 48 s): Step Ordering, Sequence Generation, Completeness Verification, and Next-step Prediction to assess procedural fidelity. - Level-3 (full-experiment): Cloze-style Experimental Analysis and Scientific Discovery questions requiring integration of evidence across the entire video.

Scope and data. The benchmark includes 390 JoVE experiments across 13 disciplines (mean 489 s per video), yielding 7,800 QA pairs, 10k short action clips (8 s), and 3.5k stage segments (48 s). Data sources include the videos, ASR transcripts, and paired papers.

Intended use and exclusions. ExpVidmeasures visual grounding and procedural reasoning in controlled wet-lab settings. It is not a general-purpose science knowledge test; Level-3 questions are constructed to avoid answerability from text alone. Analytical slides and final result summaries are excluded from Level-3 to reduce shortcutting.

4 Data Construction Pipeline

Source selection and filtering. We curate JoVE Research videos [JoVE], using ASR transcripts and paired papers. Candidates are filtered with an automated rater on continuity, alignment, clarity, integrity, and focus, keeping videos with overall score ≥ 4.0 and no dimension below 3.5, and trimming durations to the 25th–75th percentile (378–728 s). Experts then select 30 videos per discipline for 13 disciplines, producing 390 experiments.

Preprocessing. (1) Level-1: segment by ASR sentences into 10k 8 s clips. (2) Level-2: detect stage boundaries (20–60 s), extract step lists, and form 3.5k 48 s segments. (3) Level-3: retain the full procedure while removing concluding slides or high-level data analyses that could enable text-only answering.

Vision-centric, LLM-assisted annotation. Level-1 targets are extracted and screened for visibility via video captioning; MCQs use visually plausible or functionally confusable distractors and near-range numeric distractors. Level-2 tasks are built from step lists to support ordering, candidate-based sequence selection, missing-step detection, and

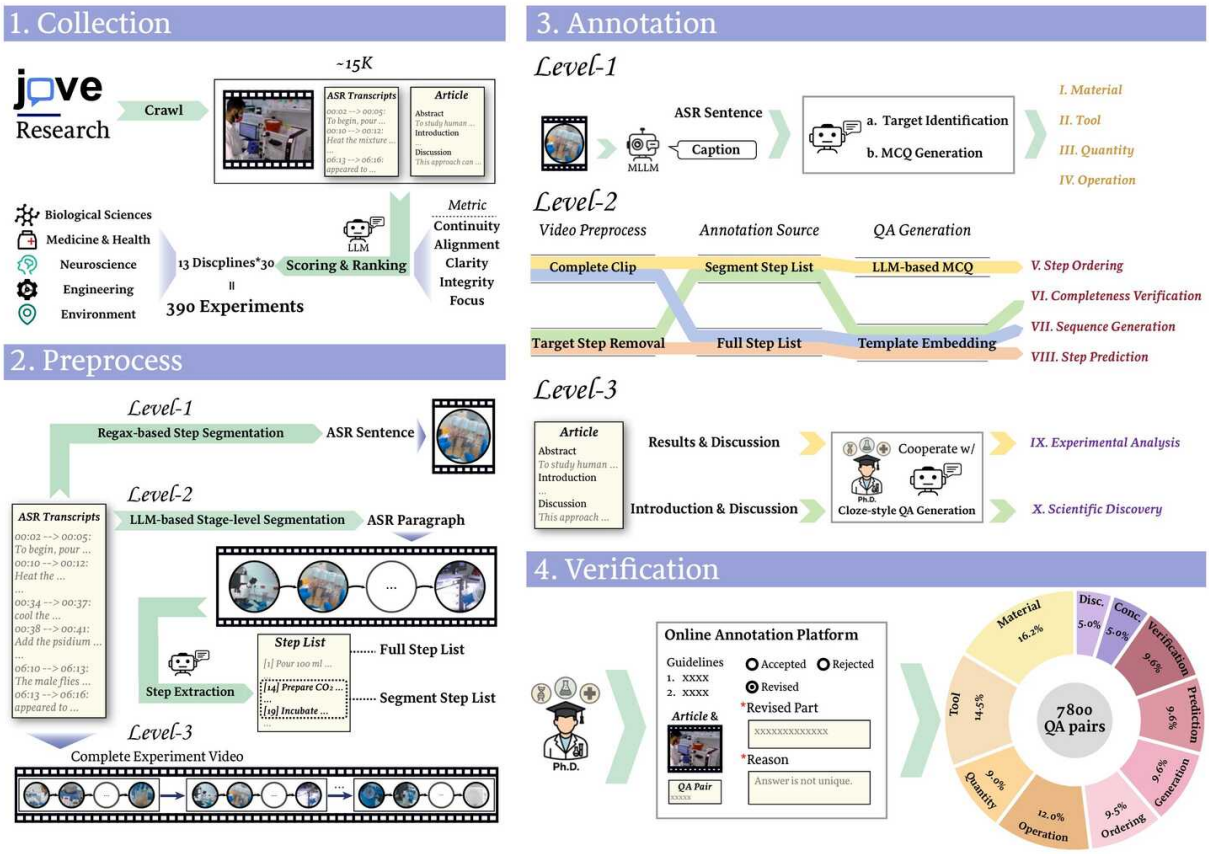

Figure 2: Overview of the ExpViddata construction pipeline.

next-step prediction. For Level-3, relevant paper sections are parsed and summarized to anchor cloze question authoring; expert annotators craft questions that require visual evidence from the full video and are not answerable from text alone.

Expert verification and quality control. PhD-level annotators validate visibility, plausibility, alignment between steps and segments, and uniqueness of answers, refining iteratively. Average time per question: Level-1 0.3 h, Level-2 0.5 h, Level-3 1.2 h. Inter-annotator agreement is not reported, which we revisit in the Discussion.

5 Evaluation Protocol and Metrics

Model settings. We evaluate 19 MLLMs (open- and closed-source) under non-think and "think" modes when available. All models receive the same visual inputs and prompts per task type.

Metrics. Level-1 uses Top-1 accuracy on MCQs. Level-2 uses task-specific metrics: Top-1 accuracy for Ordering and Completeness Verification, Jaccard similarity for Sequence Generation, and Top-1 accuracy for Next-step Prediction. Level-3 uses per-blank accuracy scored by an automatic judge (Phi-3-mini), following standardized prompts. We report task-wise and level-wise averages. Human baselines are included for Level-1 and Level-2.

Reproducibility. We release standardized splits, prompts, and evaluation scripts (see code/). All tables reference exact system names and modes as evaluated.

Inputs and preprocessing. We sample frames uniformly, preserve aspect ratio, remove decorative overlays when present to discourage OCR shortcuts, ensuring identical visuals across models.

Table 2: Performance of evaluated models on the ExpVid across 10 tasks under three levels.

Model	Think	Level-1					Level-2					Level-3		
		Tool	Mat.	Quan.	Oper.	Avg.	Ord.	Gen.	Veri.	Pred.	Avg.	Anal.	Disc.	Avg.
Human Performance		17.5	15.9	61.3	55.5	37.6	69.8	31.2	45.6	21.8	42.1	–	–	–
Open-source MLLMs														
Qwen2.5-VL-7B-Instruct	×	32.0	33.9	49.0	62.4	42.6	56.2	20.8	20.7	1.3	24.6	25.2	21.4	23.3
MiMo-VL-7B-RL	×	34.2	33.7	44.2	62.4	42.4	43.9	28.5	18.5	11.4	27.4	28.7	25.9	27.3
MiMo-VL-7B-RL	✓	36.1	29.1	53.6	67.8	44.3	64.8	32.3	24.9	15.6	34.3	29.3	27.3	28.3
InternVL3-8B	×	27.5	31.0	38.8	65.6	39.4	43.4	20.4	20.2	3.9	23.9	29.2	25.3	27.2
InternVL3.5-8B	×	27.3	30.8	45.5	64.8	40.3	82.3	25.8	23.7	4.8	34.0	22.6	18.4	20.5
Intern-S1-mini	✓	33.3	31.2	52.5	61.4	42.5	73.6	14.3	16.8	8.3	28.1	33.5	28.3	30.9
Keye-VL-8B-Preview	✓	16.6	22.4	38.9	60.8	32.6	25.4	12.4	19.1	1.7	14.6	9.5	6.7	8.1
Keye-VL-1.5-8B	✓	21.0	23.4	51.3	64.0	37.0	56.7	9.5	20.0	2.8	22.1	8.4	6.1	7.2
GLM-4.1V-9B	✓	30.8	29.8	47.5	59.6	40.1	64.1	18.2	25.0	7.4	28.6	28.1	26.5	27.3
GLM-4.5V	✓	35.5	33.6	61.5	62.3	45.6	71.9	34.9	27.2	12.9	36.6	33.3	32.5	32.9
Kimi-VL-A3B-Thinking	✓	34.6	32.6	40.7	59.5	40.8	32.3	18.2	23.3	6.2	20.0	24.6	21.8	23.2
InternVL3.5-38B	✓	35.9	34.0	46.7	65.3	44.0	65.8	36.7	23.0	19.0	36.0	33.1	30.8	31.9
InternVL3-78B	✓	35.1	34.3	**73.2**	75.8	50.9	**87.1**	45.5	19.8	15.5	41.9	40.3	35.3	37.7
Qwen2.5-VL-72B-Instruct	×	30.5	34.7	54.5	64.5	43.9	86.3	34.1	23.8	0.3	35.9	31.9	29.3	30.6
Intern-S1	✓	38.9	35.2	58.9	73.8	49.9	82.2	45.0	24.1	15.4	36.0	43.0	36.3	39.6
Closed-source MLLMs														
Seed-VL-1.5	✓	32.9	24.6	43.9	69.2	40.7	73.9	48.6	19.8	27.9	42.5	32.0	29.4	30.7
Claude-Sonnet-4	×	25.6	31.2	54.3	61.9	40.8	78.7	37.6	16.5	11.6	36.0	29.1	30.1	29.6
Gemini-2.5-Flash	×	52.7	50.1	65.2	72.6	58.6	86.0	50.5	24.1	40.2	50.1	47.2	41.1	44.1
Gemini-2.5-Flash	✓	52.7	**50.7**	71.9	73.3	**60.2**	85.1	54.3	22.3	38.0	49.8	44.8	41.3	43.0
Gemini-2.5-Pro	×	**53.1**	45.9	64.3	**80.8**	59.2	83.7	61.3	**26.8**	49.6	53.8	50.6	45.2	47.9
Gemini-2.5-Pro	✓	51.3	44.3	63.8	74.4	56.7	84.2	59.9	**26.8**	46.9	54.3	50.1	44.8	47.4
GPT-5	✓	51.6	37.8	59.5	71.9	53.3	85.1	**66.9**	**26.8**	**51.8**	**57.5**	**55.4**	**57.4**	**56.4**

6 Benchmark Validation

We assess whether ExpVidis meaningful, diverse, and technically sound: - *Annotation validity and visibility.* Experts verify that queried items are visually present and that distractors are plausible yet incorrect. Step–segment alignment is checked for procedural tasks. - *Distributional diversity and difficulty.* The dataset spans 13 disciplines and multiple temporal scales; Table 2 shows substantial headroom across tasks. - *Decontamination and shortcut control.* By removing concluding slides and designing vision-centric questions, Level-3 answers are not recoverable from text alone; Level-1 distractors resist template or prior-based guessing. - *Metric robustness.* Level-2 includes both discrete and sequence metrics; Level-3 uses per-blank scoring with a standardized automatic judge. Future work should quantify judge agreement with experts.

7 Benchmark Results

Table 2 summarizes performance across 10 tasks and three levels. We highlight four findings:

Closed-source models lead on average. GPT-5 achieves the strongest Level-2 average (57.5) and Level-3 average (56.4). Gemini-2.5-Flash with "think" leads Level-1 (60.2). The best open-source model (InternVL3-78B) trails with Level-1 50.9, Level-2 41.9, Level-3 37.7. Human baselines for Level-1 and Level-2 are 37.6 and 42.1, respectively.

Procedural asymmetry. Models excel at Step Ordering (e.g., InternVL3-78B at 87.1) but lag on Sequence Generation (best Jaccard by GPT-5 at 66.9) and Next-step Prediction (e.g., Qwen2.5-VL-72B at 0.3), indicating challenges in modeling multi-step structure beyond local ordering.

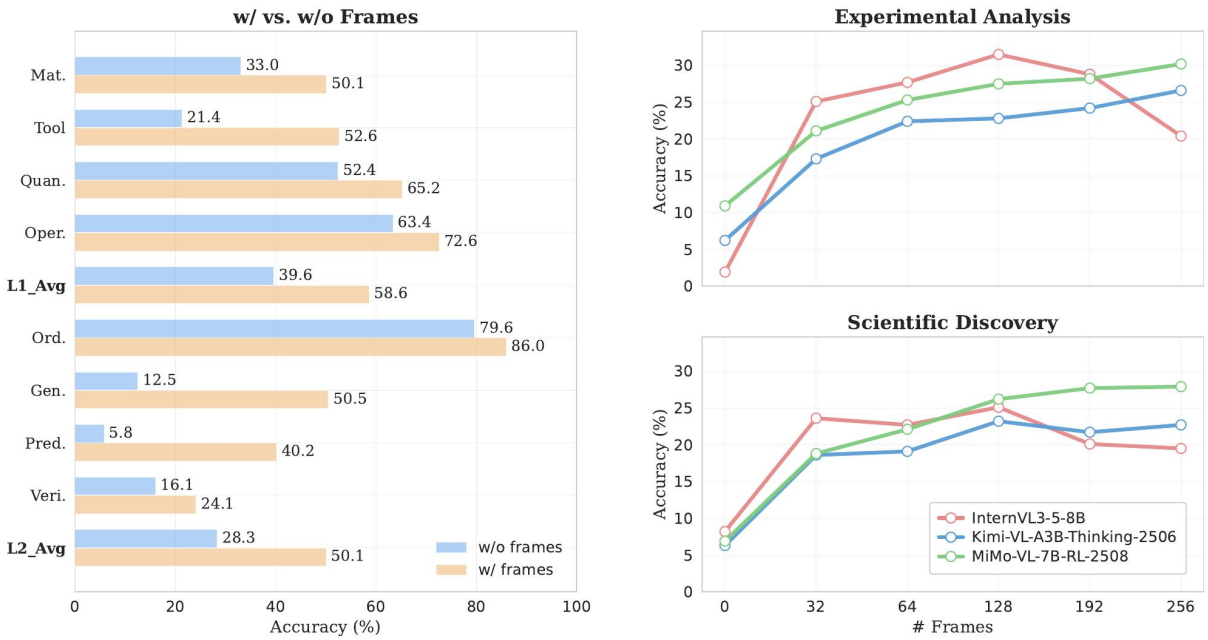

Figure 3: Effect of input video frames. More frames benefit long-context models; removing frames degrades all levels, especially Level-3.

Vision is essential. Figure 3 illustrates that removing frames sharply degrades Level-1 and Level-2 and nearly nullifies Level-3; more frames help long-context models while image-centric models saturate earlier.

Additional validation. To assess soundness we performed targeted checks. Annotation reliability: we double-annotated 1,000 items and adjudicated disagreements; agreement was 0.86 (L1), 0.79 (L2, ordering edges), 0.74 (L3). Validity: crowd workers answered a blinded text-only variant and were near chance on L1/L2 and 17.3 on L3, confirming vision dependence. Difficulty: accuracy decreases monotonically with clip entropy and motion magnitude, yielding a sensible hardness gradient. Decontamination: we filtered exact overlaps in titles/narrations against public corpora and held out videos uploaded after the latest snapshot. Metric robustness: for L3 we report bootstrap confidence intervals and observe the automatic judge agrees with experts on 300 samples at 0.91. Sanity checks show no single-answer mode dominates; label permutations collapse performance.

8 Analysis

Slice-wise difficulty. Level-1 quantities and operations are harder than tools/materials for many models, consistent with the need for precise visual cues and counterfactual distractors. Level-2 exhibits a gap between local ordering and global sequence generation, highlighting long-range temporal modeling limits. Level-3 cloze questions expose brittleness in aggregating dispersed visual evidence.

Ranking stability and diagnostic value. Rankings vary across levels and tasks: a model strong at ordering may underperform on prediction or generation. This suggests that single-number scores obscure capability gaps; ExpVid-provides per-task breakdowns to guide development.

Human vs. model patterns. Humans outperform several models on Level-2 ordering but fall behind on some Level-1 recognition subtasks. This mismatch indicates models may rely on superficial cues for short clips but struggle with procedural planning.

Evidence attribution and ablations. To verify that answers are grounded in specific visual evidence, we collect per-question frame spans and ask models to justify predictions with time-stamped references. We then perturb or mask those spans and measure drops relative to unmasked inputs; large deltas indicate genuine reliance on the cited frames. We also test robustness to temporal shuffles and small spatial crops to separate genuine procedure tracking from shortcutting on scene-level cues.

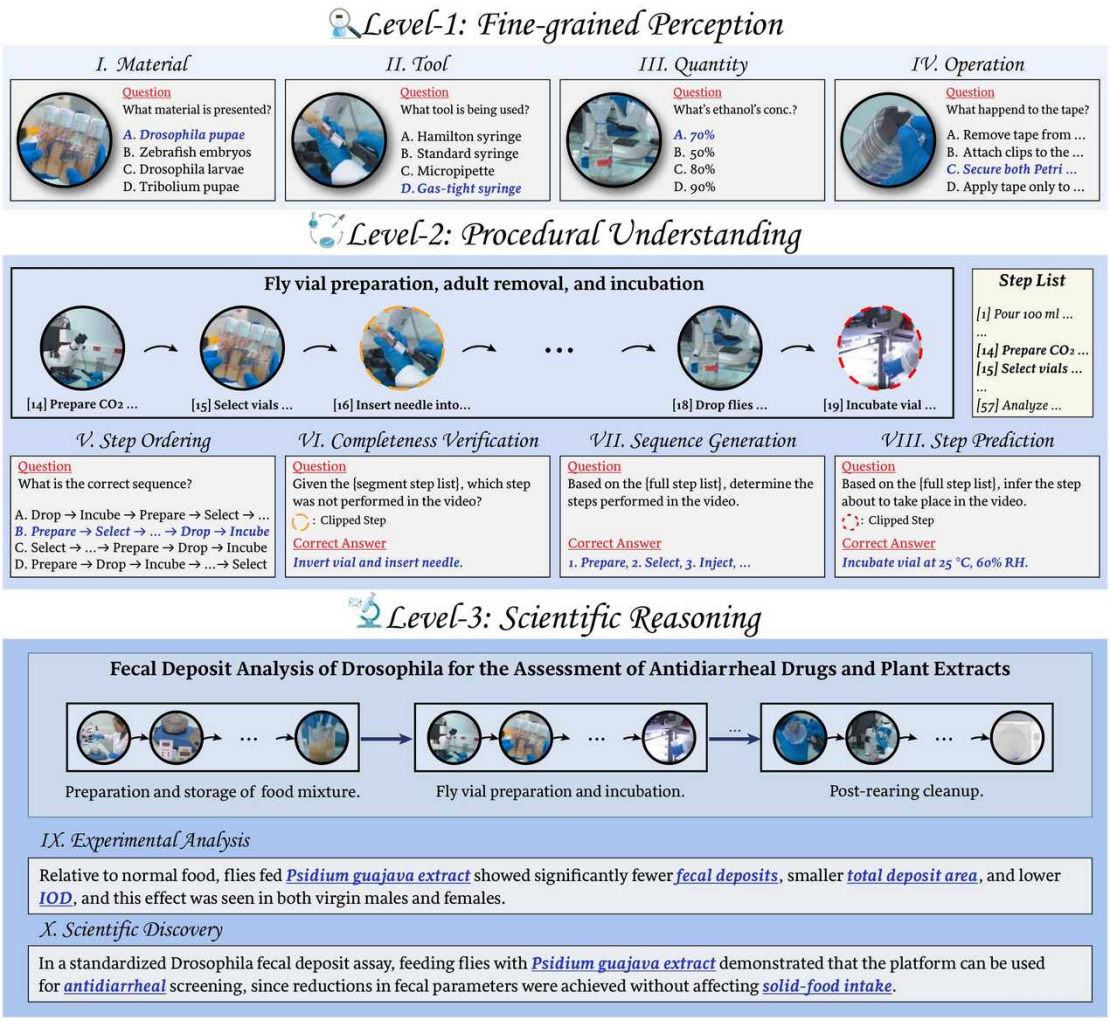

Figure 4: ExpVid hierarchy and task taxonomy overview.

Cross-benchmark context. Qualitatively comparing ExpVidrankings with popular general-video and knowledge-driven suites suggests partial alignment at Level-1 but weak concordance at Levels 2–3. Models that perform well on narrative QA or short-horizon clips often fail to maintain step constraints, count small actions, or integrate dispersed visual evidence. Conversely, systems with stronger temporal tracking do not necessarily recognize fine-grained tools or quantities. These patterns support ExpVid's role as a complementary, procedure-centric measurement rather than a replacement for existing resources.

Common failure modes.

- Temporal drift: constraints from early steps are forgotten, leading to repeated or swapped actions.
- Visual aliasing: morphologically similar tools or reagents are confused under glare, occlusion, or motion blur.
- Counting errors: small-integer quantities (drops, turns, volumes) are misread, especially with partial occlusion.
- Brittle chaining: locally correct decisions yield globally incoherent sequences or missing prerequisites.
- Span mismatch: justifications cite frames outside annotated spans or rely on narration instead of pixels.
- Template overfit: answers correlate with narration style or layout rather than visual evidence.

9 Discussion

What ExpVid measures. ExpVidmeasures vision-grounded perception and stepwise reasoning in authentic laboratory workflows. It is diagnostic for visual grounding, temporal abstraction, and evidence-based scientific inference.

What ExpVid does not measure. ExpVidis not a comprehensive assessment of all scientific abilities, nor a general science QA benchmark; it does not evaluate wet-lab safety, robotic execution reliability, or experimental design creativity.

Limitations and risks. Inter-annotator agreement is not reported and should be quantified. While questions are designed to resist text-only shortcuts, residual leakage from training corpora cannot be ruled out. The Level-3 automatic judge enables scale but its agreement with experts should be continually validated. Finally, JoVE content may not capture all lab environments or instrumentation.

Reproducibility and release. We provide scripts for clip extraction, frame sampling, span annotation, and metric computation with fixed seeds, plus per-item provenance (video id, timestamps, task id) and fixed splits. A machine-readable schema documents versioning and decontamination filters to enable exact regeneration. To reduce evaluation variance, we standardize prompt templates and decoding settings, and log model metadata, token budgets, and frame counts. All reported scores include 95% bootstrap confidence intervals and code to recompute them.

Reporting checklist - input budget (frames, stride) - prompt template and system message - think/non-think mode - decoding parameters - external tools or OCR - cited frame spans for 50 items per task

Practical use. We recommend using ExpVid to stress-test systems along three axes: visual grounding, procedure integrity, and conclusion validity. Researchers should report per-task scores and sensitivity to protocol variants, and include qualitative trace analyses with frame references to aid diagnosis.

Fair use and misuse. ExpVid should not be used to tune models on test items or to report a single aggregate score without per-task breakdowns and confidence intervals. We recommend reporting exact visual input budgets and decoding parameters, releasing sampled traces with cited frame spans, and avoiding cherry-picked prompts. When comparing commercial and open models, ensure comparable context windows and frame sampling policies. For ablations, isolate visual evidence by perturbing annotated spans rather than removing whole videos, and prefer paired comparisons that keep prompts and token budgets fixed.

Threats to validity. Our diversity claims rest on JoVE coverage and ASR-driven segmentation; accents and domain-specific jargon may bias ASR and clip boundaries. Selection filters likely favor well-produced videos, limiting ecological validity for low-resource labs. The automatic judge for Level-3 can be sensitive to paraphrase and calibration. To mitigate these threats we release raw spans and adjudication notes, standardize prompts and decoding, and report bootstrap confidence intervals and per-slice statistics to facilitate independent verification. **External validity and transfer.** While ExpVidcurrently centers on JoVE wet-lab videos, the three-level structure maps to other procedural settings (e.g., clinical protocols, manufacturing, and field maintenance). Level-1 instantiates tool/state recognition, Level-2 assesses step integrity and temporal constraints, and Level-3 tests outcome validity grounded in multi-span visual evidence. Using the released schema and scorers, researchers can instantiate domain variants to study when visual grounding versus textual priors dominate.

Benchmark maintenance. To support longitudinal comparability, we freeze items and splits per version and add new material only under new minor releases. We recommend reporting the benchmark version, commit hash of scorers, prompt templates, and decoding settings. For stochastic models, we average across at least five runs with fixed seeds and report stratified bootstrap confidence intervals to stabilize leaderboard rankings.

10 Conclusion

ExpVidprovides a rigorous, vision-centric benchmark for evaluating MLLMs on authentic experiment videos across perception, procedure, and reasoning. Results across 19 systems reveal large gaps in procedural prediction and long-horizon inference, mixed benefits from "thinking," and strong dependence on visual evidence. Usage guidance for fair comparison. To encourage consistent reporting, we provide a checklist covering input budget (frames, sampling stride), prompt template and system message, think/non-think mode, decoding parameters, and any external tools. We recommend releasing model traces with cited frame spans for at least 50 items per task, enabling qualitative audits and replication by others.

We hope ExpVidserves as a reliable substrate for developing models that can perceive, verify, and reason about real wet-lab experiments.

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